

Benchmarking VeSpA, Segment Anything, YOLOv8, and a Hybrid VeSpA-SAM Approach for Vessel Segmentation in Histological Images

Repository-based technical manuscript

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Abstract

This manuscript documents the segmentation benchmark implemented in the repository for vessel analysis in histological images with DAB immunostaining. Four approaches were compared under a unified binary-mask interface: a custom VeSpA image-processing pipeline (color-based), the Segment Anything Model (SAM) with automatic mask generation, a pretrained Ultralytics YOLOv8 segmentation model, and a hybrid VeSpA-SAM approach combining color priors with prompt-guided foundation-model refinement. Evaluation was performed on four histological TIFF images with paired binary vessel masks using Dice coefficient, intersection over union, precision, and recall. In the current repository benchmark, VeSpA achieved the highest mean Dice coefficient (0.7433), followed by the hybrid VeSpA-SAM pipeline (0.6979), SAM (0.2988), and YOLOv8-seg (0.1263). Mean runtime per image was 2.18 s for VeSpA, 81.87 s for the hybrid method, 53.28 s for SAM, and 2.19 s for YOLOv8-seg. This manuscript reports the implemented methods and the benchmark outcomes produced by the repository code.

1 Introduction

The Spatial Vessel Analysis project website describes a reproducible workflow for extracting vessel-related features from histological images, with the broader aim of characterizing vascular conformation, remodeling, and spatial relationships to drug distribution. In this context, vessel segmentation is a core preprocessing step because the fidelity of downstream morphometric and spatial analyses depends directly on the quality of the binary vessel mask.

In practical settings, candidate segmentation methods span classical image-processing pipelines, promptable foundation models, and general-purpose neural instance segmentation architectures. These model families differ substantially in their inductive biases, computational cost, and expected transfer performance when applied to histological vascular structures without task-specific retraining.

This repository implements a controlled benchmark of four segmentation approaches on a small image-and-mask dataset stored in TIFF and PNG formats. The compared methods are: (i) a custom VeSpA pipeline implemented as a deterministic color-space and morphology-based segmentation workflow, (ii) the Segment Anything Model (SAM) using automatic mask generation [1, 2], (iii) an Ultralytics YOLOv8 segmentation model using the pretrained `yolov8n-seg.pt` weights [3], and (iv) a *novel hybrid VeSpA-SAM approach* that combines the color-based domain knowledge of VeSpA with the segmentation refinement capabilities of SAM using prompt-guided prediction. The hybrid method represents a new strategy for integrating classical domain priors with foundation models in histology.

The objective of this manuscript is not to establish general biological or clinical performance, but to document the repository implementation, standardize evaluation across methods, and report the measured performance obtained from the current codebase. All reported numerical results are taken directly from the benchmark outputs generated by the repository and no additional experiments were introduced beyond those artifacts.

2 Methods

2.1 Dataset and evaluation protocol

The benchmark iterates over all images stored in `datasets/images` and matches each image with a binary ground-truth mask in `datasets/groundtruth` using the naming rule `<image_stem>_mask.png`. In the present repository state, the dataset contains four histological TIFF images: `ROI_1.tiff`, `ROI_2.tiff`, `ROI_3.tiff`, and `ROI_4.tiff`. Three images have spatial resolution 4001×4001 pixels and one image has resolution 4000×4000 pixels. Ground-truth masks are binary PNG files with values $\{0, 255\}$.

The benchmark supports three ground-truth categories: `category_1_observers`, which evaluates against individual observer masks from SLR and GG; `category_2_intersection`, which evaluates against the pixelwise intersection of both observers; and `category_3_union`, which evaluates against the pixelwise union of both observers. When `category_1_observers` is selected, the benchmark runs separately for each observer and records the observer name in the output metrics.

All methods expose a unified interface,

```
run_model(image_path) -> np.ndarray
```

which returns a binary mask of type `uint8`. The benchmark resizes each predicted mask to the corresponding ground-truth resolution using nearest-neighbor interpolation to ensure pixelwise comparability. Reproducibility is promoted by fixed random seeds (`seed=42`) for Python, NumPy, and PyTorch where applicable.

2.2 VeSpA pipeline

The VeSpA implementation is imported from `pipelines/VeSpA/Segmentation+Measurements.py`. The pipeline reads each color image in BGR format, converts it to a CMYK representation, and uses the yellow channel as the main contrast source for vessel extraction. The yellow channel is smoothed by Gaussian filtering and binarized with Otsu thresholding. The resulting binary image is refined through morphological closing, dilation, and erosion, followed by hole filling and cleanup operations using closing and opening. The final binary vessel mask is returned as the segmentation output. This pipeline is deterministic and does not rely on learned weights.

2.3 SAM pipeline

The SAM implementation is provided in `pipelines/sam/run_sam.py`. The current benchmark uses the ViT-B SAM checkpoint stored in `models/sam_vit_b_01ec64.pth`. Input images are converted from BGR to RGB and downsampled to a maximum side length of 1024 pixels before inference in order to control memory use and execution time. Automatic mask generation is then applied using the repository configuration `points_per_side=16`, `pred_iou_thresh=0.8`, `stability_score_thresh=0.9`, `crop_n_layers=0`, and `min_mask_region_area=64`. Candidate

masks covering more than 25% of the resized image area are filtered out to suppress large background regions. The remaining masks are merged by union and resized back to the original image resolution.

2.4 YOLOv8 segmentation pipeline

The YOLO implementation is provided in `pipelines/yolo/run_yolo.py`. The benchmark loads the pretrained Ultralytics segmentation model from `models/yolov8n-seg.pt`. Each image is processed with inference size `imgsz=1024` and `retina_masks=True`. When the model returns multiple instance masks, each mask is resized to the original image resolution with nearest-neighbor interpolation and the final binary prediction is obtained as the pixelwise union of all instances.

2.5 Hybrid VeSpA-SAM pipeline

The hybrid implementation is provided in `pipelines/hybrid/run-vespa_sam.py`. This method combines the stain-sensitive color prior used by VeSpA with the promptable `SamPredictor` interface from SAM. Each image is converted from BGR to CMYK, the yellow channel is extracted, and Otsu thresholding plus morphological closing are used to obtain a binary stain mask representing candidate endothelial regions. Connected-component analysis is then applied to this stain mask to derive one positive centroid prompt and one bounding box per retained component.

The RGB image is resized to a maximum side length of 1024 pixels before SAM inference, and all prompt coordinates and boxes are scaled accordingly. For each stain-derived component, the hybrid pipeline calls `SamPredictor.predict` with a positive point prompt and the corresponding component box. SAM returns a component-specific mask, and the implementation normalizes mask dimensionality, resizes each component mask back to the original image size, and combines the refined masks by logical union. Final morphological opening and closing are applied to suppress small artifacts and fill local gaps. In contrast to the automatic-mask SAM baseline, this hybrid approach constrains segmentation to stain-supported candidate regions while still leveraging SAM for boundary refinement.

2.6 Evaluation metrics

Four binary segmentation metrics are implemented in `evaluation/metrics.py`: Dice coefficient, intersection over union (IoU), precision, and recall. Let P denote the predicted foreground set and G the ground-truth foreground set. The metrics are computed as

$$\begin{aligned} \text{Dice} &= \frac{2|P \cap G|}{2|P \cap G| + |P \setminus G| + |G \setminus P|}, \\ \text{IoU} &= \frac{|P \cap G|}{|P \cap G| + |P \setminus G| + |G \setminus P|}, \\ \text{Precision} &= \frac{|P \cap G|}{|P \cap G| + |P \setminus G|}, \quad \text{Recall} = \frac{|P \cap G|}{|P \cap G| + |G \setminus P|}. \end{aligned}$$

When the denominator of a metric is zero, the implementation returns 1.0.

2.7 Outputs generated by the benchmark

The benchmark writes per-image and per-model metrics to `results/metrics/results_all_categories.csv`, model-level means to category-specific summary files (`results/metrics/summary_cat1_SLR.`

csv, etc.), and mean runtime per image to category-specific runtime files (`results/metrics/runtime_summary_cat1_SLR.csv`, etc.). For qualitative inspection, the pipeline also saves binary predictions and color overlays in `results/prediction/`. Summary visualizations consist of a Dice boxplot, a mean-IoU bar chart, and a four-metric comparison plot saved under `results/metrics/`.

3 Results

Table 1 reports the mean segmentation performance across the four benchmark histological images for all four methods. VeSpA achieved the best mean Dice coefficient (0.7433) and IoU (0.5929). The hybrid VeSpA-SAM method ranked second with a mean Dice coefficient of 0.6979 and the highest mean precision (0.8739), indicating that stain-guided prompting substantially reduced false positives relative to the generic SAM baseline. However, the hybrid method remained slightly below VeSpA in overlap-based metrics because its recall (0.5958) was lower than VeSpA’s 0.7052.

Table 1: Mean performance metrics computed from `results/metrics/results_all_categories.csv`.

Model	Dice	IoU	Precision	Recall
VeSpA	0.7433	0.5929	0.8059	0.7052
VeSpA+SAM (Hybrid)	0.6979	0.5397	0.8739	0.5958
SAM	0.2988	0.1782	0.6403	0.1968
YOLOv8-seg	0.1263	0.0711	0.7557	0.0736

Per-image results are shown in Table 2. The hybrid method produced substantial gains over automatic SAM on all four images, raising Dice from 0.3388 to 0.5952 on ROI_1, from 0.1607 to 0.7229 on ROI_2, from 0.3459 to 0.7247 on ROI_3, and from 0.3497 to 0.7487 on ROI_4. Despite these improvements, VeSpA remained the top-performing method on every image in this benchmark.

Table 2: Per-image Dice and IoU values averaged across all ground truth categories from `results/metrics/results_all_categories.csv`.

Image	Model	Dice	IoU
ROI.1	VeSpA	0.6845	0.5204
ROI.1	VeSpA+SAM (Hybrid)	0.5952	0.4244
ROI.1	SAM	0.3388	0.2041
ROI.1	YOLOv8-seg	0.0736	0.0383
ROI.2	VeSpA	0.7442	0.5931
ROI.2	VeSpA+SAM (Hybrid)	0.7229	0.5662
ROI.2	SAM	0.1607	0.0874
ROI.2	YOLOv8-seg	0.0512	0.0263
ROI.3	VeSpA	0.7694	0.6253
ROI.3	VeSpA+SAM (Hybrid)	0.7247	0.5692
ROI.3	SAM	0.3459	0.2093
ROI.3	YOLOv8-seg	0.0706	0.0366
ROI.4	VeSpA	0.7750	0.6327
ROI.4	VeSpA+SAM (Hybrid)	0.7487	0.5991
ROI.4	SAM	0.3497	0.2120
ROI.4	YOLOv8-seg	0.3097	0.1833

Runtime measurements are summarized in Table 3. VeSpA and YOLOv8-seg were essentially tied for the fastest method, at 2.18 s and 2.19 s per image respectively. The hybrid VeSpA-SAM method required 81.87 s per image on average, making it the slowest configuration in the current repository benchmark because it runs prompt-guided SAM refinement over multiple stain-derived components per image; this is more than 1.5 times the runtime of automatic SAM (53.28 s), reflecting the added cost of stain-guided prompt generation on top of SAM inference.

Table 3: Mean runtime per image, averaged across the four ground truth categories (`results/metrics/runtime_summary_cat1_GG.csv`, `runtime_summary_cat1_SLR.csv`, `runtime_summary_cat2.csv`, `runtime_summary_cat3.csv`).

Model	Runtime (s/image)
VeSpA	2.18
YOLOv8-seg	2.19
SAM	53.28
VeSpA+SAM (Hybrid)	81.87

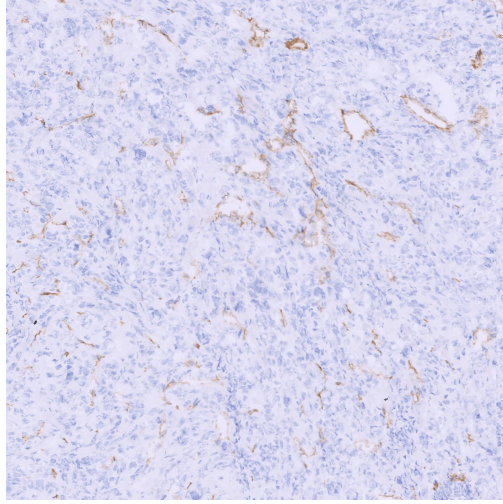
Figure 2 visualizes the distribution of Dice scores, and Figure 3 shows the mean IoU comparison. Figure 4 additionally breaks down Dice, IoU, precision, and recall side by side for all four methods. The plots confirm that the hybrid pipeline closes much of the gap between baseline SAM and VeSpA while clearly outperforming YOLOv8-seg in overlap-based metrics, and that the hybrid method attains the highest precision of any method at the cost of lower recall. Qualitative overlays saved under `results/prediction/` show that the hybrid method is more conservative than VeSpA, with higher precision but lower recall on several images.

Figure 1 provides a representative qualitative example for ROI_1. Panel (a) shows the raw histological image. Panels (b)–(e) show the overlays generated by the benchmark for VeSpA, VeSpA+SAM (Hybrid), SAM, and YOLOv8-seg, respectively.

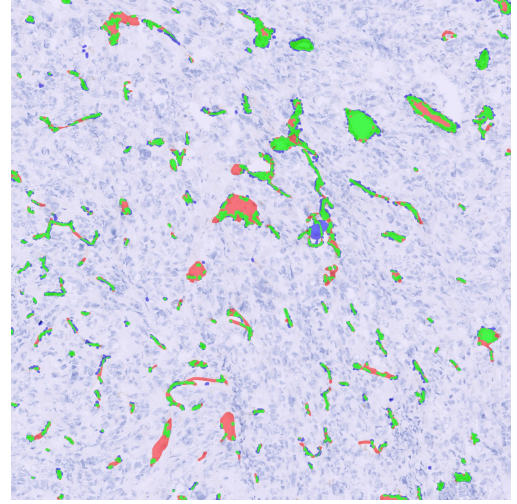
4 Conclusion

Within the current repository benchmark, the custom VeSpA pipeline remained the best-performing method for vessel segmentation in the available histological images, achieving the highest mean Dice (0.7433) and IoU (0.5929) while remaining computationally efficient at 2.18 s per image, essentially tied with YOLOv8-seg for the fastest method. The hybrid VeSpA-SAM pipeline ranked second with a mean Dice of 0.6979 and the highest mean precision (0.8739), showing that stain-guided prompting can recover much of the performance gap between generic SAM and a domain-tuned classical pipeline. However, the hybrid method was slower than both VeSpA and automatic SAM, requiring 81.87 s per image on average in the current implementation.

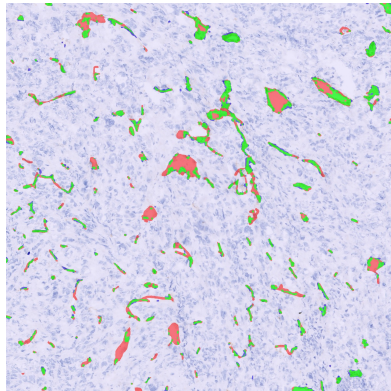
These findings should be interpreted within the scope of the implemented experiment. The dataset contains four images, and neither SAM nor YOLOv8-seg was fine-tuned for histological vessel segmentation in this repository. Accordingly, the reported results demonstrate comparative behavior of the current codebase rather than a general ranking for these model families. Future work could reduce the hybrid runtime by pruning prompts, merging nearby components before SAM refinement, or batching prompts more efficiently; additional work could also extend the benchmark with larger curated histology datasets and supervised fine-tuning of YOLO segmentation on task-matched annotations. In the broader Spatial Vessel Analysis framework described on the project website, improved segmentation quality would support more reliable downstream vessel statistics and spatial analyses.



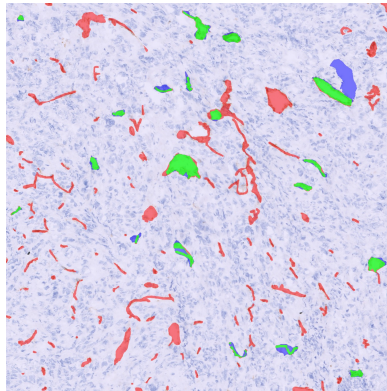
(a) Raw image



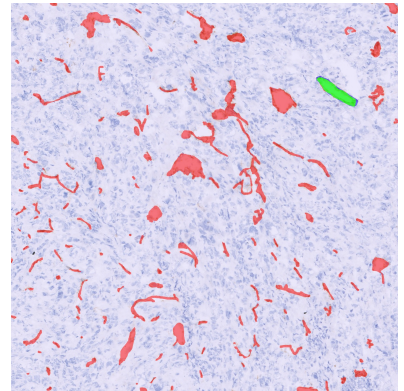
(b) VeSpA overlay



(c) VeSpA+SAM (Hybrid) overlay



(d) SAM overlay



(e) YOLOv8-seg overlay

Figure 1: Representative qualitative example for ROI_1. Panel (a) shows the raw histological image. Panels (b)–(e) show prediction overlays for VeSpA, VeSpA+SAM (Hybrid), SAM, and YOLOv8-seg. In the overlay visualizations, green denotes ground-truth vessel pixels, red denotes predicted vessel pixels, and yellow denotes spatial overlap between prediction and ground truth.

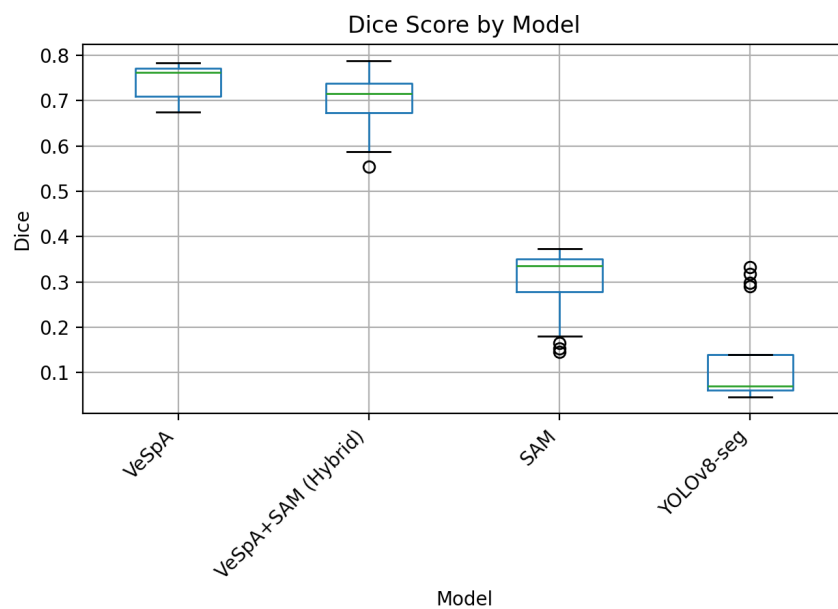


Figure 2: Distribution of Dice scores across the four benchmark histological images.

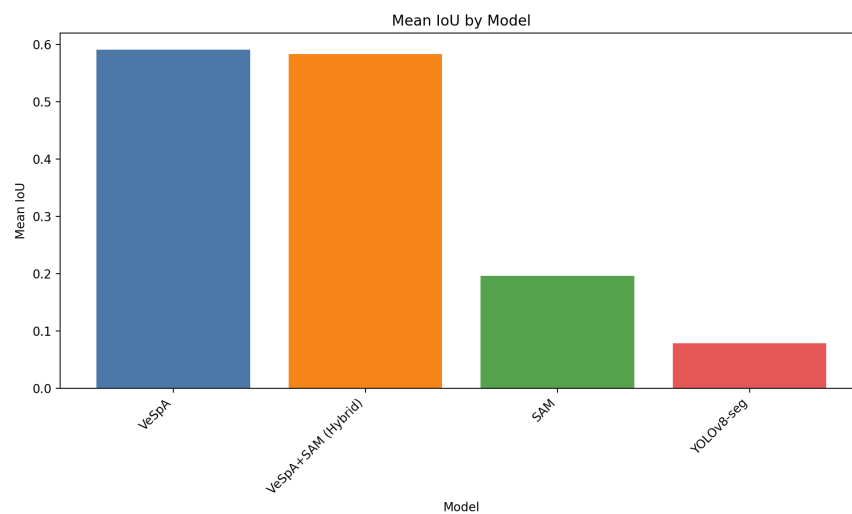


Figure 3: Mean IoU comparison across the four evaluated segmentation methods.

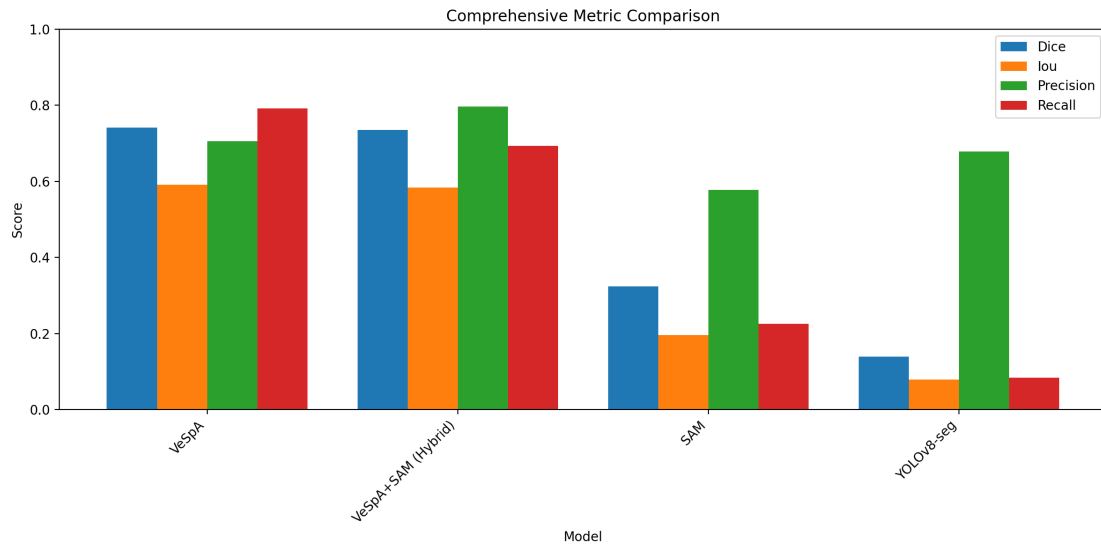


Figure 4: Comprehensive comparison of Dice, IoU, precision, and recall across the four evaluated segmentation methods.

References

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